

The background of the slide features a large, faint, orange-colored watermark of the Tsinghua University seal. The seal is circular and contains the university's name in both Chinese and English, along with the founding year 1896.

信息论基础

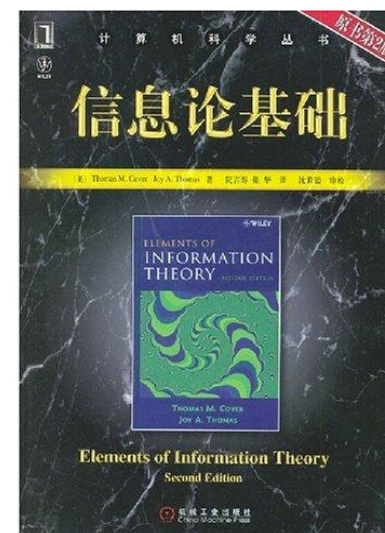
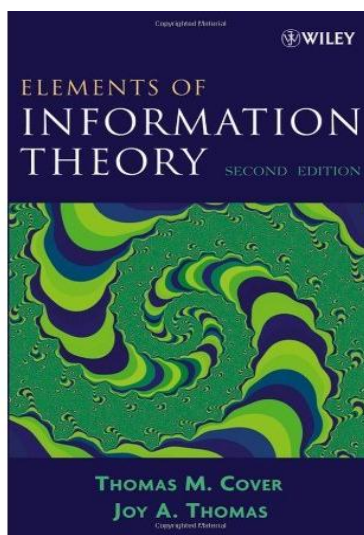
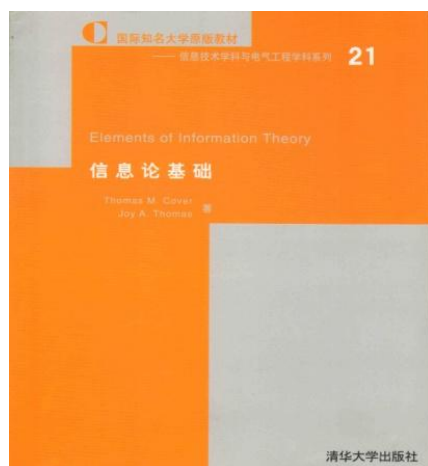
Elements of Information Theory

王 慧 明
Hui-Ming Wang



课程信息 Course Information

- ☐ Period : 40 class hours/16 weeks
- ☐ Credits: 2.5 units
- ☐ Text book
 - Elements of Information Theory, Thomas M. Cover and Joy A. Thomas (*first/second edition*, Chinese edition)





Course Information

☐ Prerequisite courses (先修课程)

■ Probability theory (概率论)

☐ Random variable, probability density function (pdf), mathematical expectation, variance, correlation,

☐ 随机变量、概率密度函数, 数学期望, 方差, 相关

■ Stochastic process (随机过程)

☐ Stationary, ergodic, auto- correlation

☐ 平稳, 各态历经, 自相关

■ Fundamental communication theory (通信原理)

☐ Digital communications (数字通信)

☐ source coding (信源编码)

☐ channel coding (信道编码)

☐ Modulation (调制)



Course Information

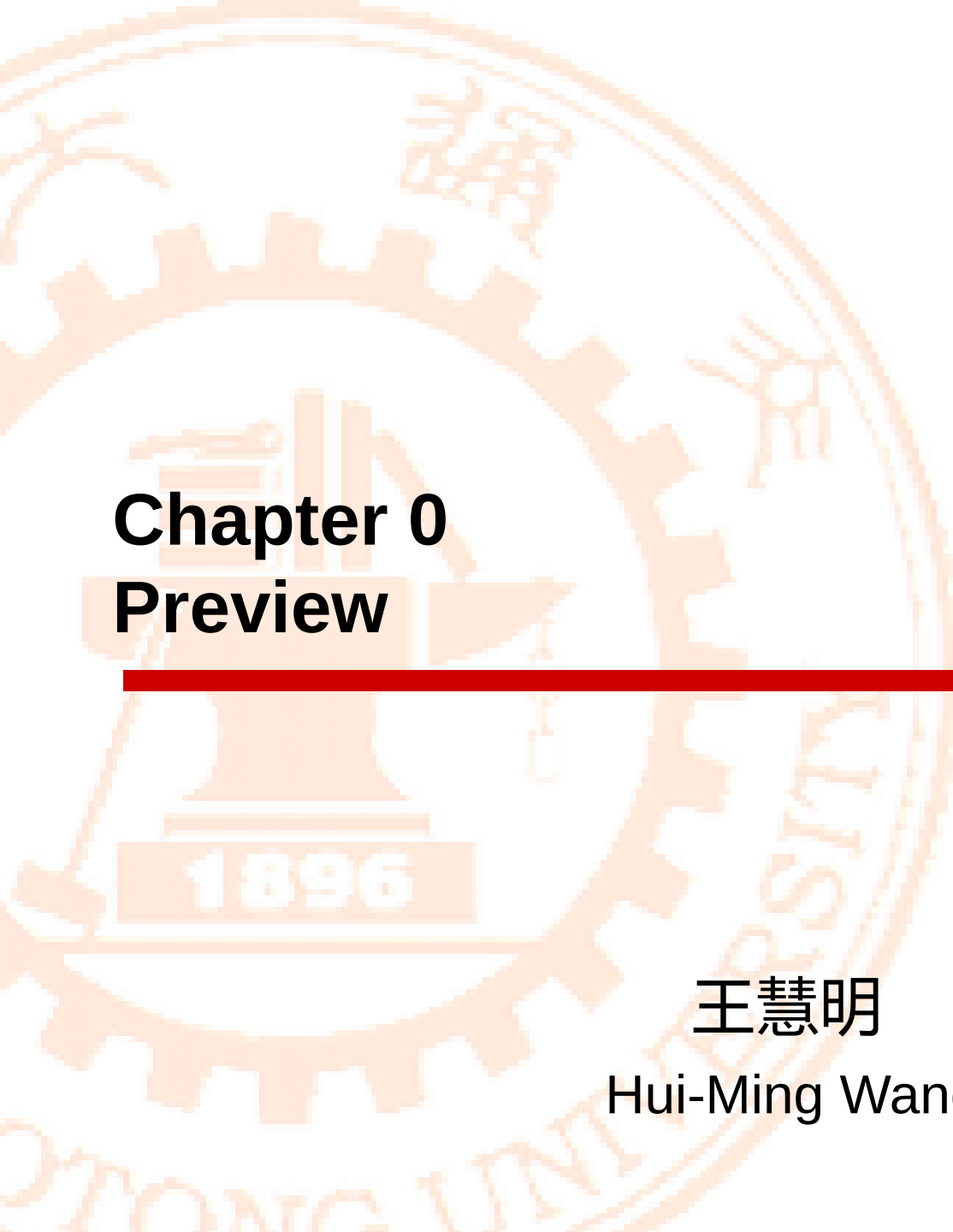
- Homeworks and exam
 - 3 homework, 1 test
 - Final exam
- Hui-Ming Wang (王慧明)
 - 教授、博导、教育部青年CJ学者
 - 彭康楼207室, Email: 4372667@qq.com
 - Website: <http://gr.xjtu.edu.cn/web/hmwang>

研究方向：无线通信、智能通信

B5G/6G关键技术及应用

- 基于机器学习和人工智能的无线通信
- 智能反射表面/通信感知一体化
- 无线通信物理层安全与内生安全



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Chapter 0

Preview

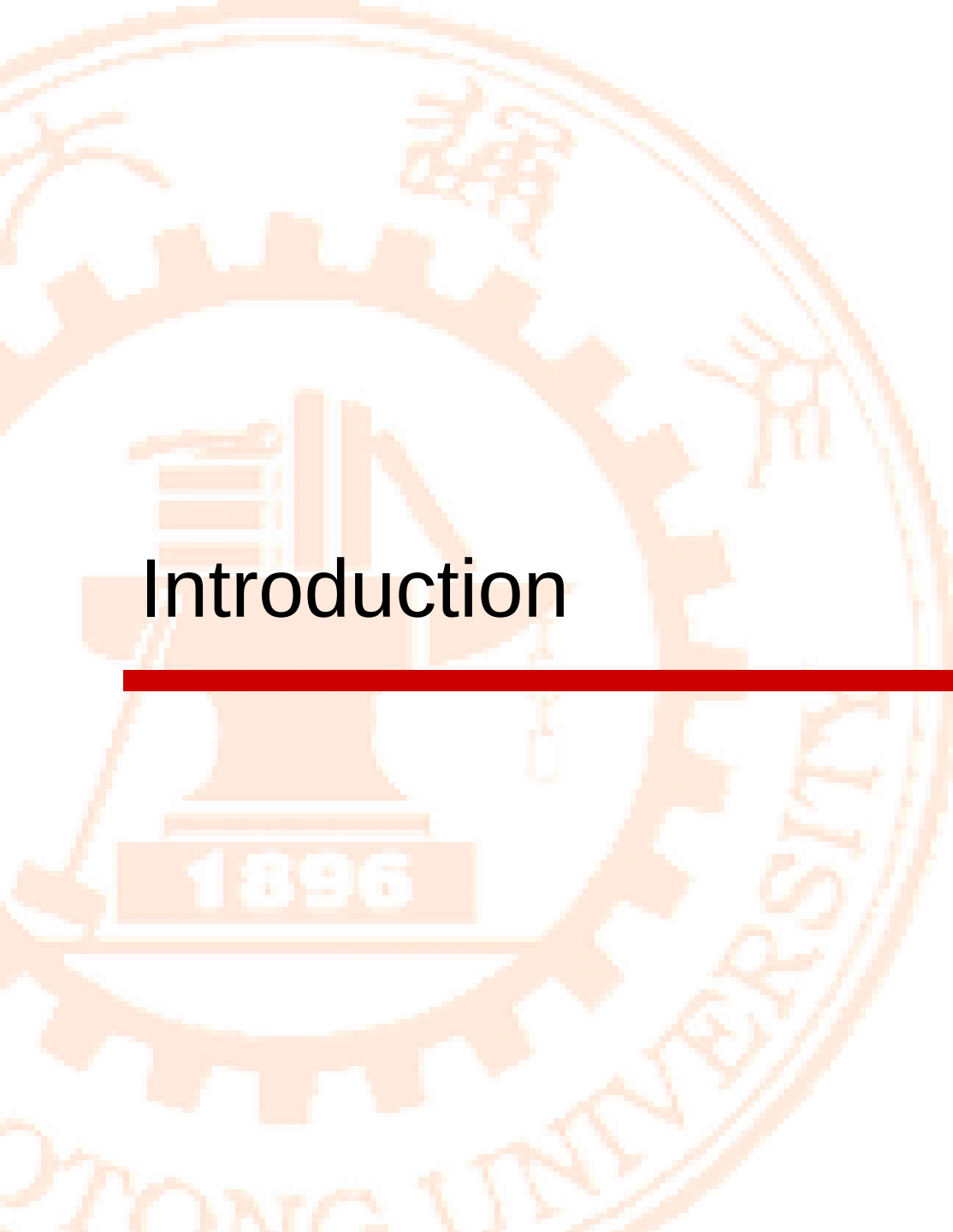
王慧明

Hui-Ming Wang



Outline

- ☐ Introduction
- ☐ Course Outlines (课程大纲)
- ☐ Digital Communications

The background of the slide features a large, faint, orange-toned seal of Tsinghua University. The seal is circular, with a gear-like outer border. Inside the border, the university's name is written in Chinese characters at the top and 'TSINGHUA UNIVERSITY' in English at the bottom. The center of the seal depicts a traditional Chinese building with a flag on top, and the year '1896' is inscribed at the bottom of the central emblem.

Introduction



What is information?

□ OED（牛津字典）says:

- **facts** provided or learned about something or someone.
- **what is conveyed or represented** by a particular arrangement or sequence of things.

□ Websters（韦伯大辞典）says:

- the communication or reception of **knowledge or intelligence**
- the attribute inherent in and communicated by one of two or more alternative sequences or arrangements of something that produce specific effects.
- **something (as a message, experimental data, or a picture)** which justifies change in a construct that represents physical or mental experience or another construct.

□ 汉语解释

- 信息是对客观世界中各种事物的运动状态和变化的反映，是客观事物之间相互联系和相互作用的表征，表现的是客观事物运动状态和变化的实质内容。



What is information?

□ 消息

- 2020年的新冠肺炎爆发
- 语言含义。消息承载信息。

□ 信号

- 摔杯为号、使了一个眼色
- 声信号、电信号、光信号、电磁波信号，信号承载消息

□ 信息

- A quantitative measure of the content of information; specially : a numerical quantity that measures the uncertainty in the outcome of an experiment to be performed

用数值来量化度量某个事件的“不确定性”的多少。

What is information?

- Such information has “semantic meaning”.

We don't mean semantics, we want a “quantifiable ” meaning.

信息论不关注语义（语言含义），关注的是如何“量化度量”信息的多少（不确定性的多少）

- Key is communication, relaying a message from someone to someone else.

跟通信密切相关，消息从一端到另一端的传递

我在跟你通信，那么我给你传递了多少信息？如何量化这个信息的多少？





What is information theory?

- 本课程所涉及的信息论是**狭义信息论**
 - 主要创始人是美国科学家香农 (C. E. Shannon) (1916-2001)
 - 1940年 Doctor of MIT
 - 在1948年和1949年先后发表的两篇文章，奠定了信息论的基础
 - “The mathematical theory of communication”
 - “Communication in the presence of noise”,
 - 讨论了信息的度量、信息特征、信息传输速率、信道容量、干扰对信息传输的影响等问题，全部理论建立在信息是可以度量的基础上。
-



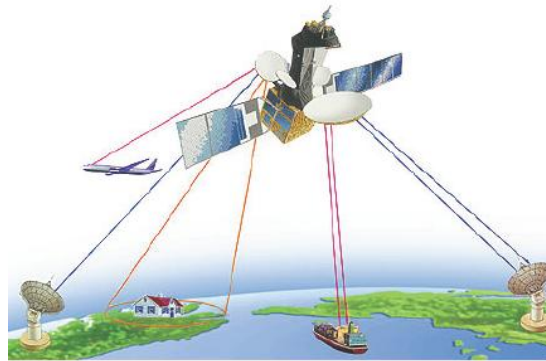
What is information theory?

Science of
transmission
compression,
and **storage** of information

信息论是运用概率论与数理统计的方法研究信息、信息熵、通信系统、数据传输、数据压缩、密码学等问题的应用数学学科。它是二十世纪中期诞生的“三大论”之一，即**控制论（维纳）**、**系统论与信息论（香农）**。经过半个多世纪的发展，信息论已经成为现代信息社会的奠基石。

Two fundamental questions (两大基本问题)

- ❑ Everyone has a mobile phone (smart phone). We may download many files and application software from the internet. The download speed is more and more fast today. So whether we can download a high definition movie in one second in the future?
- ❑ The files we download every day from the Internet have been compressed. Shall we compress a high definition movie (2GB) into 100KB in the future?





Introduction

- ❑ Origin – Communications practice and theory
- ❑ Morse 1837
 - Telegraph (wired) and Morse code
- ❑ Maxwell 1865 (electromagnetic)
 - Maxwell equations
- ❑ Bell 1876
 - Telephone (wired)
- ❑ Hertz 1887
 - Show the existence of electromagnetic waves by experiments.
- ❑ Marconi 1901
 - Radio transmission across the Atlantic.
- ❑ Around 1910's
 - Radio telegraph
- ❑ 1927-1929
 - Shortwave Television

Before the coming of the information theory, there are already many communications systems



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通信四位奠基人



詹姆斯·克拉克·
麦克斯韦

1831-1879

“电磁学”之父

1865



海因里希·鲁道夫·
赫兹

1857-1894

“电磁波”之父

1887

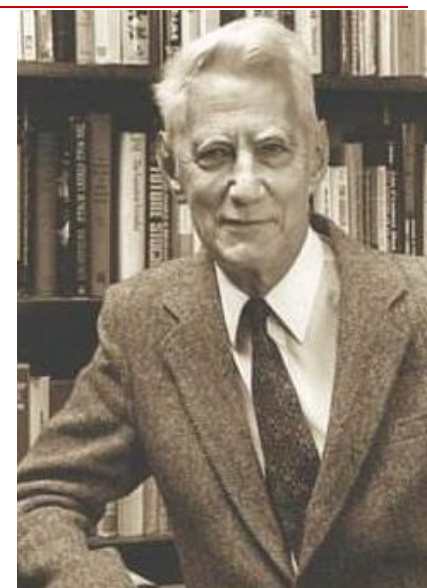


伽利尔摩·
马可尼

1874-1937

“无线电”之父

1901



克劳德·艾尔伍德·
香农

1916-2001

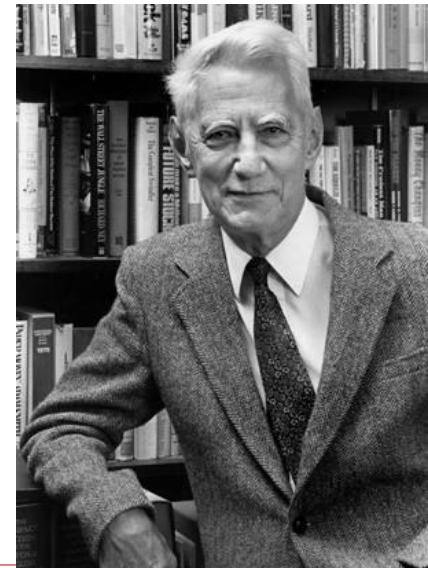
“信息论”之父

1949

Introduction

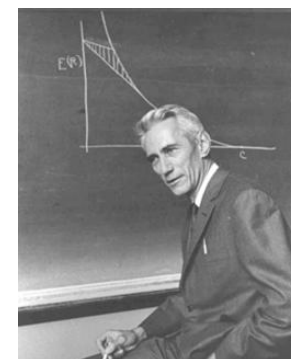
□ *Claude E. Shannon, 1916-2001*

- American mathematician, electronic engineer, and cryptographer. Known as “the father of information theory”.
- Graduated in 1936 from University of Michigan with two [bachelor's degrees](#), one in [electrical engineering](#) and the other in [mathematics](#).
- Got his master's degree from M.I.T. in 1937
- Got his [PhD](#) in [mathematics](#) from M.I.T. in 1940
- Joined the **Bell Labs**.
- Spent the rest of his life in Bell Labs and M.I.T.
- Found Information Theory in 1948





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Claude E. Shannon
statue in MIT





Introduction

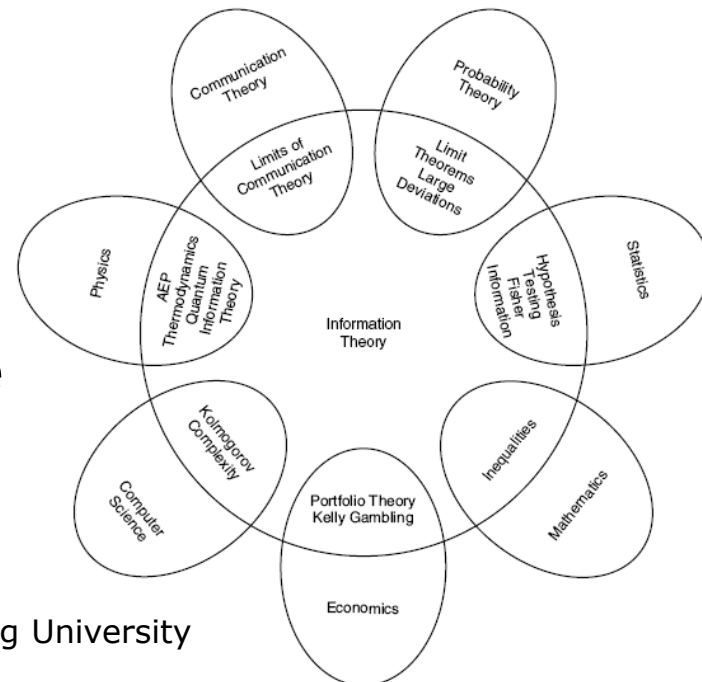
□ Information Theory

- Claude E. Shannon “A Mathematical **Theory of Communication**” in *The Bell System Technical Journal*, 1948
- The central paradigm of classical information theory is the engineering problem of ***the transmission of information over a noisy channel***. (有噪信道的信息传输问题)
- Focus on the ***quantification of information***. (信息的量化)
 - The term “information theory” is a bit too broad.
- Shannon found the fundamental ***quantification limits*** (量化的极限) on signal processing operations such as ***compressing data*** and on ***reliably storing and communicating data***.
- (Soviet Union)
Kolmogorov

Introduction

- More than Communications
 - Mathematics (Probability theory and statistics)
 - Physics (Thermo-dynamics)
 - Electrical engineering (Communications)
 - Computer science (Kolmogorov complexity)
 - **Artificial intelligence (AI)**
 - Economics

Of course, we only focus on the aspects of communications in this course.



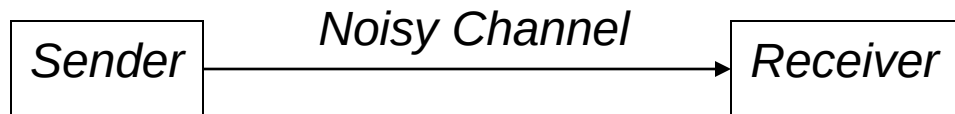


Introduction

- Varies of communications systems
 - Telegraph
 - Telephone
 - Television (cable)/Radio broadcasting
 - ADSL / VDSL
 - Cellular / mobile
 - WLAN / WiFi
 - WPAN / Bluetooth
 - **Satellite / GPS**

Introduction

□ Information theory in **Communications**

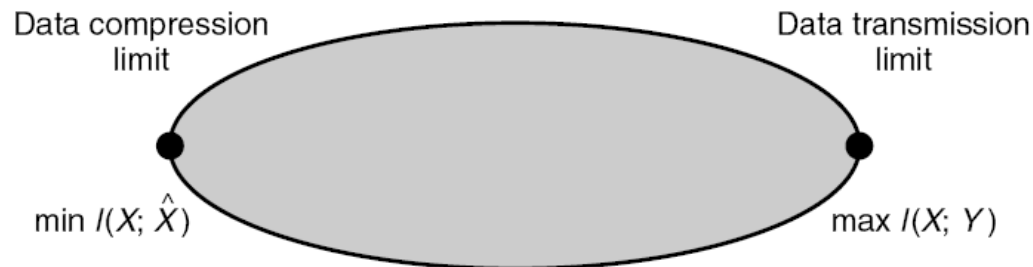


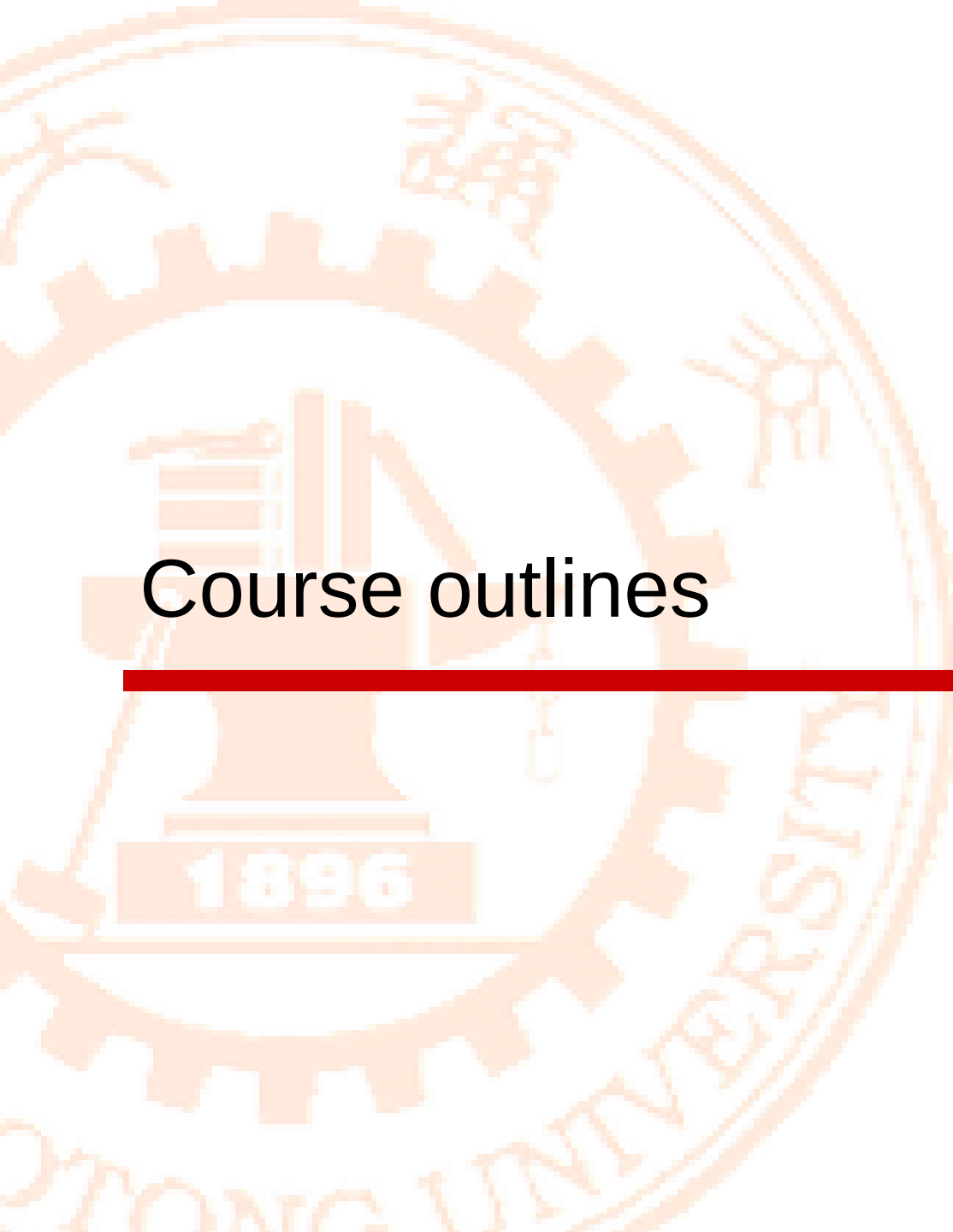
□ Two fundamental questions in communication theory

- How much a source can be compressed?
- How much information can be sent through a given channel?
 - Before, more power, more faster communications.
- How to define information?

□ Represents the extreme points of all possible schemes

- Point-point commun.
- Hard to achieve
- Practical reasons
- Code design



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Course outlines



Course outlines

- General conception(基本概念)
 - Definitions, Notions, and Math foundations
- Compression（压缩）
 - Theory
 - Coding scheme (Simple examples)
- Transmission（传输）
 - Theoretical limits
 - Coding scheme → Channel Coding (Advanced Courses)

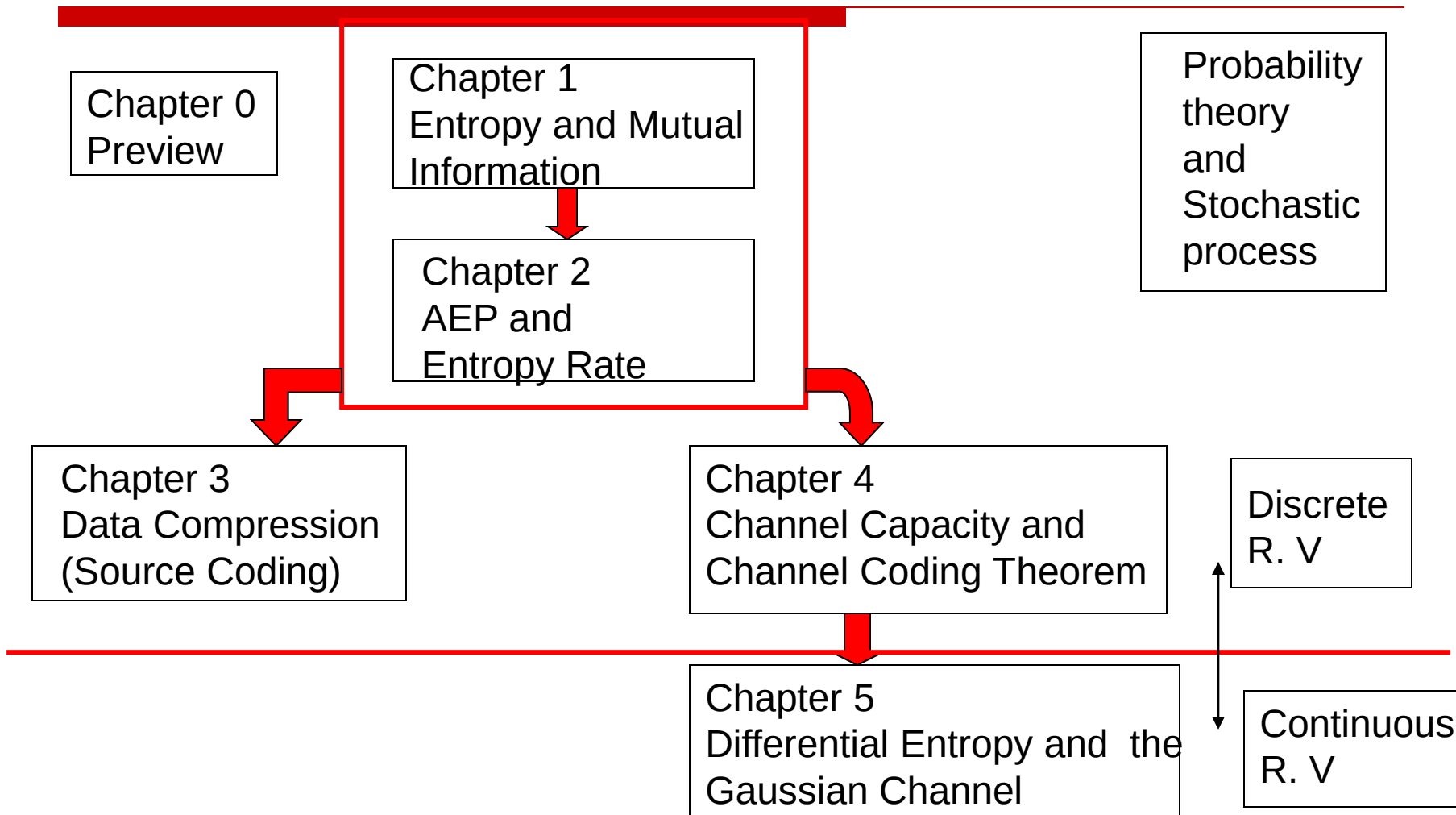


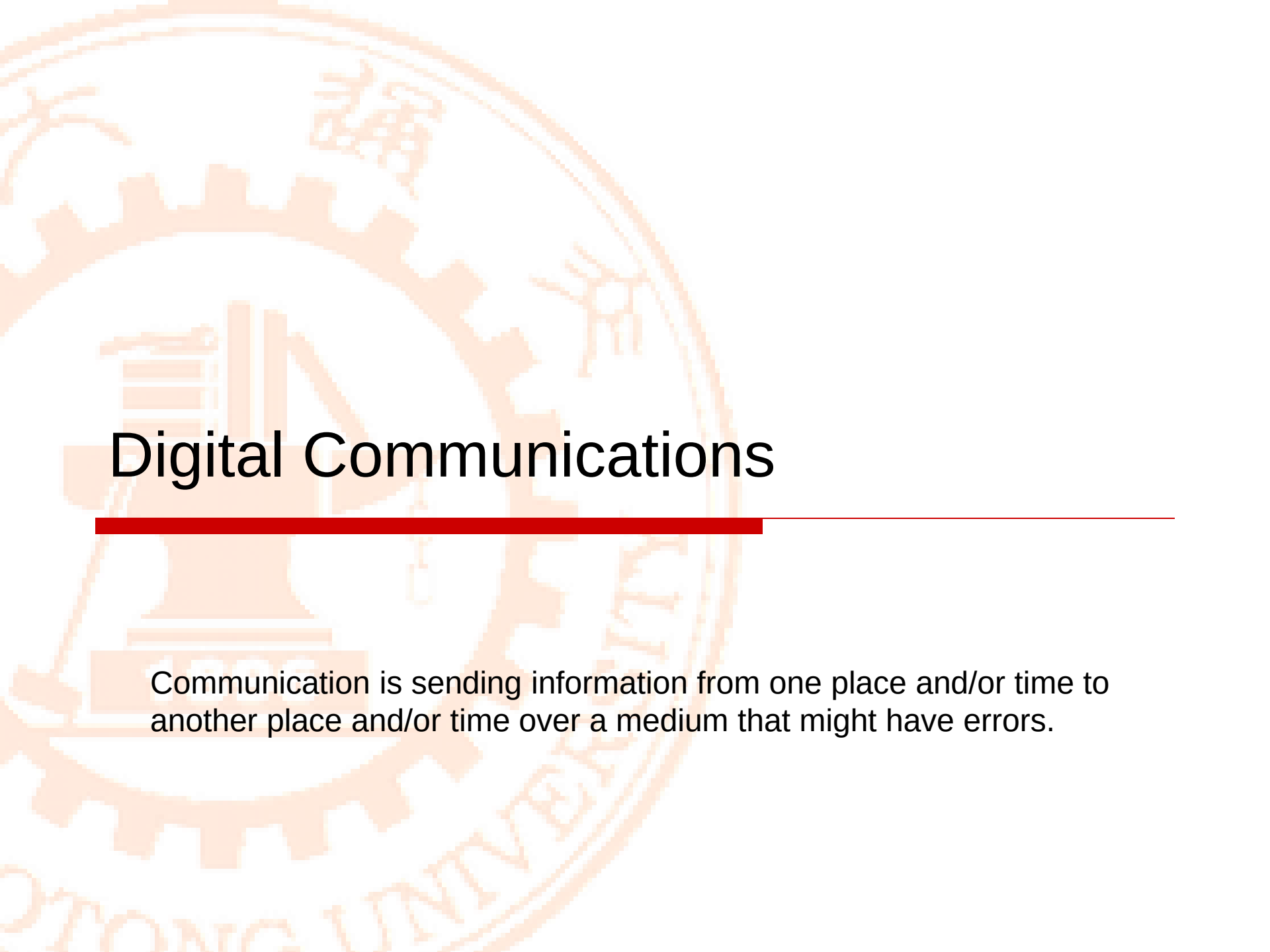
Chapters

- Chapter 0
 - Preview
- Chapter 1
 - Entropy and Mutual Information (熵和互信息)
- Chapter 2
 - Asymptotic Equipartition Property (AEP) and Entropy Rate (渐进等分性质和熵率)
- Chapter 3
 - Data Compression (Source Coding)
- Chapter 4
 - Channel Capacity and Channel Coding Theorem
- Chapter 5
 - Differential Entropy (差熵) and the Gaussian Channel



Structure



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Digital Communications

Communication is sending information from one place and/or time to another place and/or time over a medium that might have errors.

Digital Communications

□ Analog Communications

- Radio broadcast system: AM, FM
- Television systems (Now most of them are digital)
- 1G cellular/mobile system



□ Digital Communications

- 2G + Cellular/mobile (3G/4G/5G)
- WLAN/Wi-Fi
-



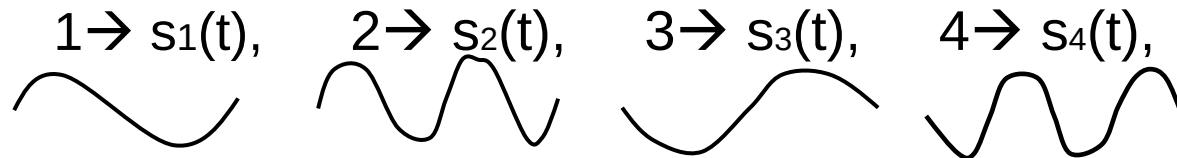
Digital Communications

□ Analog vs. Digital Communications

- Analog Commun.: Analog waveforms directly transmitted through channel. (模拟波形传输) → **Infinite possibilities**

- Digital Commun.: Using a finite waveform → **finite possibilities**

- Example: source generates messages {1,2,3,4}

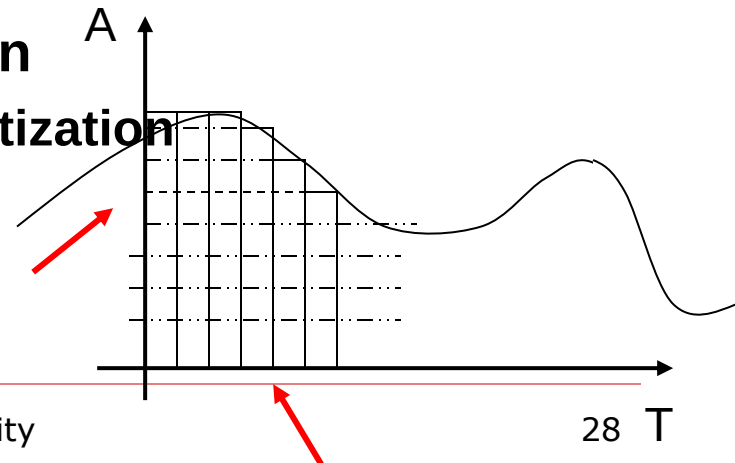


- Analog signal → Digital signal:

- Sampling – Time discretization

- Quantizing – Amplitude discretization

- { Digital signals have quantizing error
- { Digital signals are robust to noise





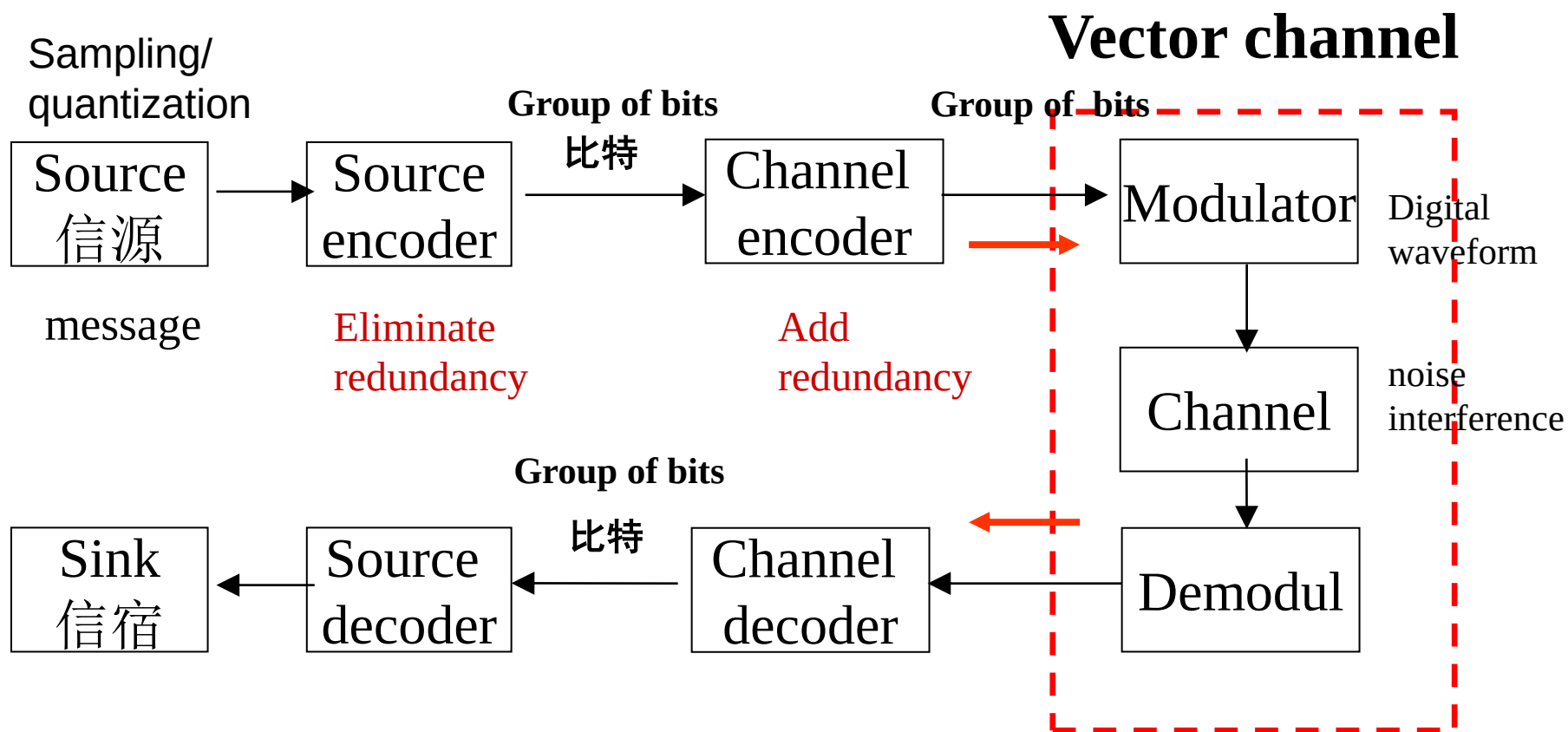
Digital Communications

- Contrary to the analog commun, we are able to **control the amount of distortion** by quantizing more or less.
- We can apply tools of digital commu. (channel coding) to guarantee perfect commun. of the quantized message if the quality of the channel is beyond a threshold (capacity).
- Information theory shows that in order to obtain the best possible performance for transmission of analog source over noisy channel, we can perform:
 - Optimal sampling (采样)
 - Optimum quantization (量化)
 - Optimum coding (编码)

Due to the above reasons, in the following, we assume the sources are all ***digital***.

Digital Communications

□ Digital communications system





Digital Communications

- Source
 - Digital: Outputs are in a set of messages. (e.g. 1 2 3... W)
 - Analog: Sampling and quantization. (采样与量化)
- Source encoder
 - Maps the vector of source symbols into a vector of bits/symbols so that the number of required bits or symbols is minimized.
 - **Eliminate as much redundancy from the source as possible.**
- Channel encoder
 - **Introduce controlled redundancy to protect from errors in vector channel.**
- Under many conditions, for single to single communications, separation between the source and channel coding is optimal.



Source coding

- Source encoder
 - Compression
 - Character -- ASCII, GB2312, GBK
 - Speech -- PCM, MP3 (MPEG1 Layer3)
 - Image -- BMP, JPG, AVI(divx, xvid, H.264), RMVB
 - How much?
- Lossless compression (无损压缩)

$$R_{compression} \left(\frac{bits}{source\ symbols} \right) \geq H_{source}$$

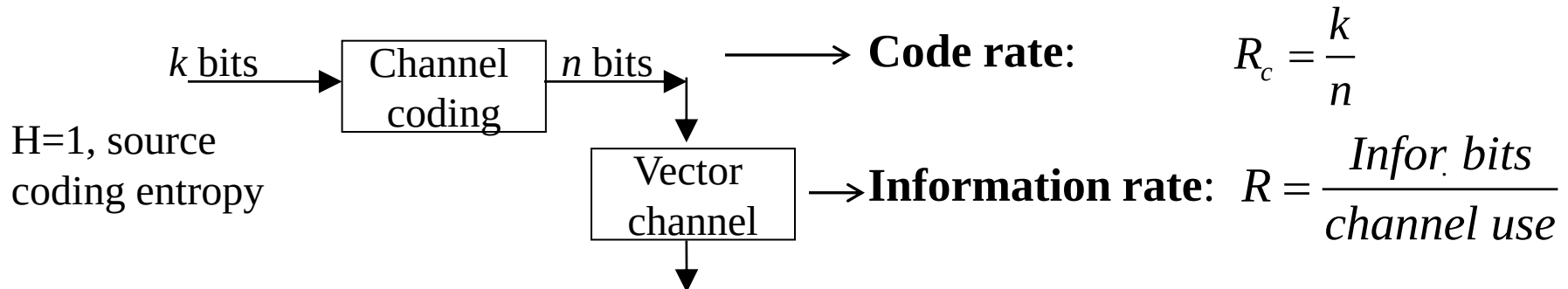
- H_{source} is the entropy of the source and depends on the statistics of the source
- Entropy is a measure of the uncertainty.
- Lossy (有损压缩)
 - A certain distortion is allowed → need to define distortion criteria.



Channel coding

□ Channel coding

- Source has been already compressed as much as possible.
- Add redundancy to protect against noise in channel.



- **Noisy channel coding theorem: Shannon**
- For any channel, we can define parameter C (capacity) so that for any fixed $R < C$, it is possible to find a sequence of codes of length n , so that $n \rightarrow \infty, P_e \rightarrow 0$



Channel coding

- Some examples
 - 1950 Hamming block code
 - 1960 Reed Solomon code
 - 1967 Viterbi decoding
 - 1993 Turbo code
 - 1963 Low-density parity-check code (LDPC)
 - 1996 rediscovered (efficient decoding algorithm)

OK, Let's Start ! 
